

Project Abstract - OnFix

Abstract: Finding a reliable and available vehicle mechanic during an emergency can often be a stressful and time-consuming process for drivers. OnFix is an on-demand mechanic application designed to solve this problem by providing a quick, reliable, and user-friendly platform that seamlessly connects drivers with nearby mechanics. The application features a mobile interface built using Flutter, while a robust Spring Boot backend paired with Redis handles fast, real-time location-based services and efficient data management.

Key features of OnFix include real-time WebSocket notifications that instantly alert mechanics of new service requests, a live emergency request system for immediate roadside assistance, and an interactive map for tracking mechanic locations. By automating the search for mechanics and facilitating fast communication, OnFix aims to bridge the gap between stranded drivers and professional repair services, ensuring that automotive help is highly accessible, transparent, and prompt.

Keywords: Application Development, Mobile Application, Flutter, Spring Boot, WebSocket, Real-Time Notifications, Location-Based Services, On-Demand Service, Automotive Assistance.